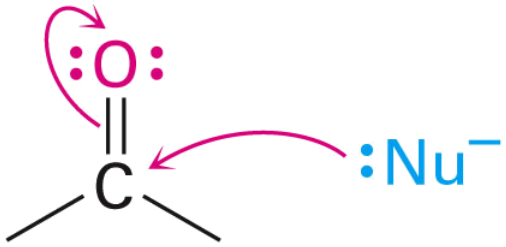
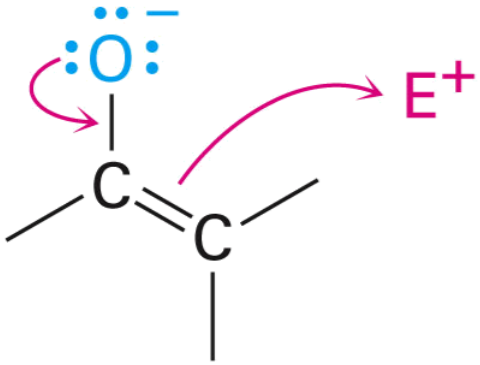
Chapter 23
Carbonyl Condensation Reactions

Condensation Reactions

- Carbonyl compounds are both the electrophile and nucleophile in carbonyl condensation reactions

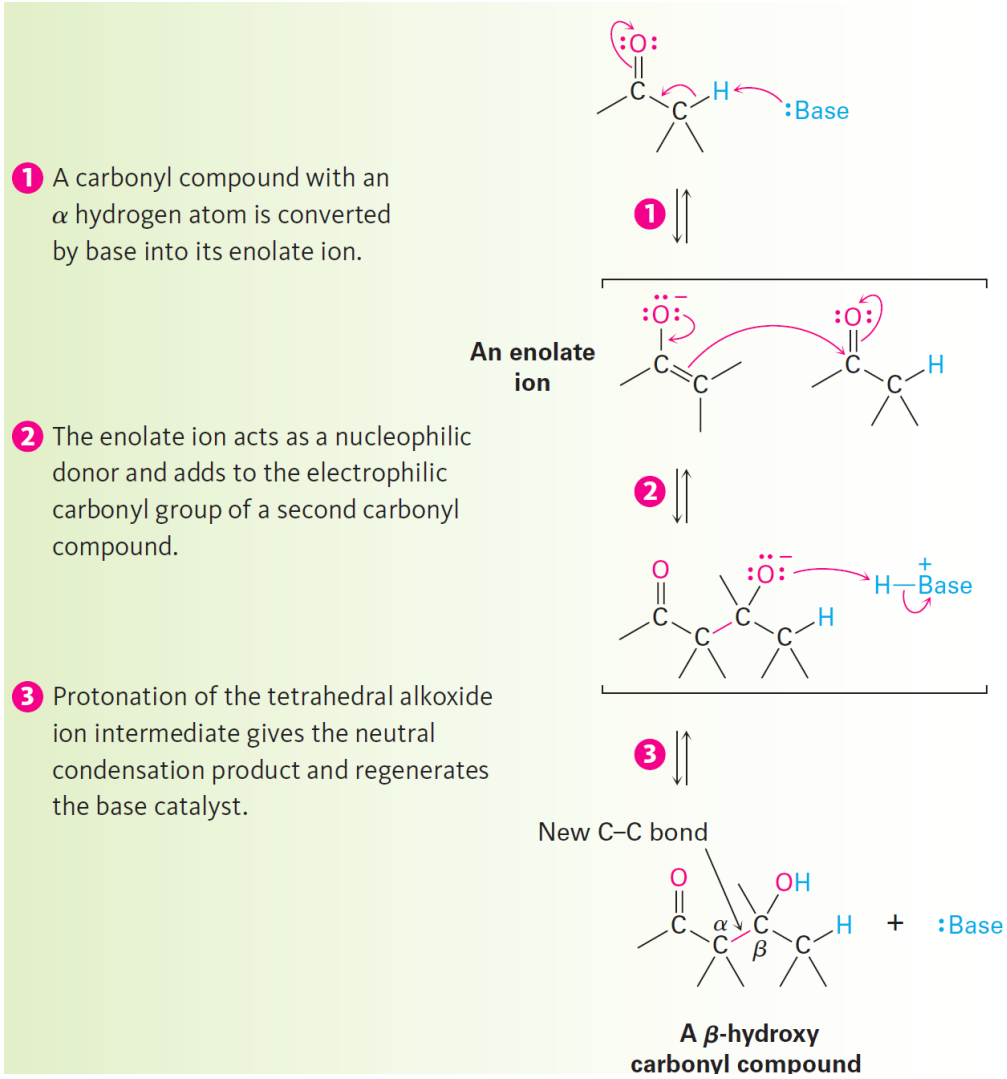


Electrophilic carbonyl group reacts with nucleophiles.



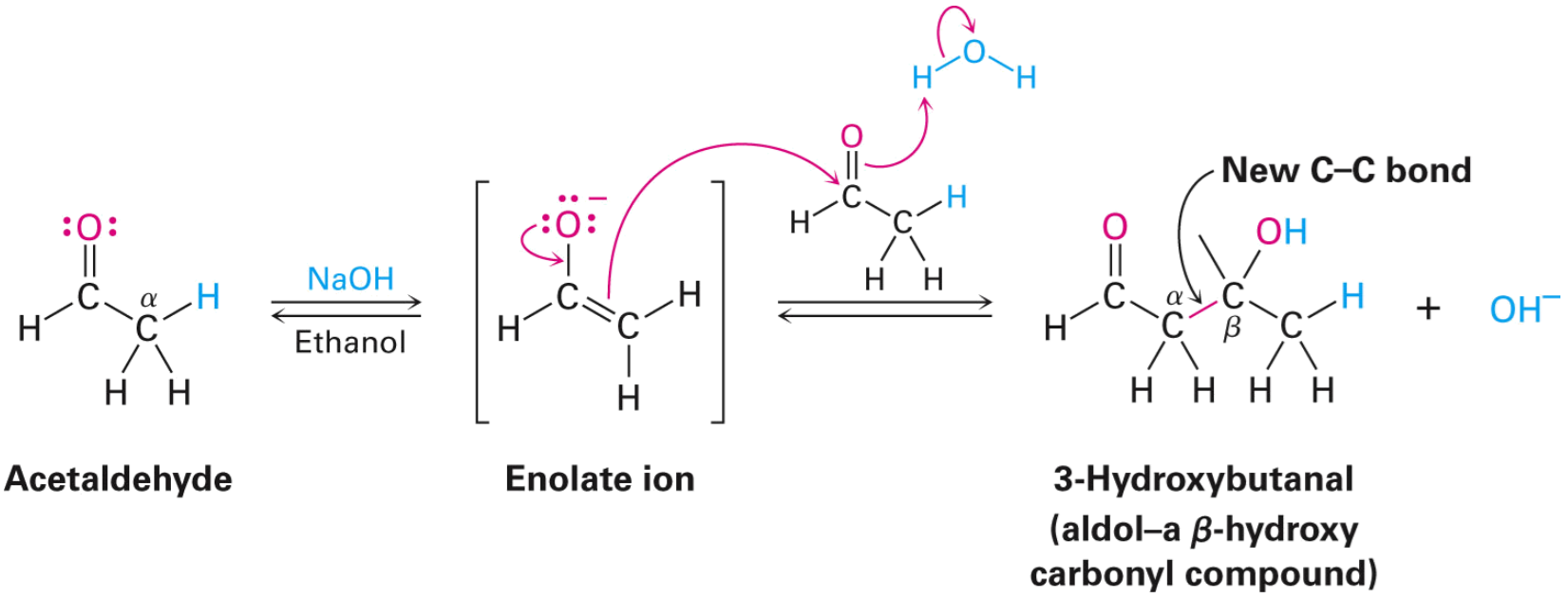
Nucleophilic enolate ion reacts with electrophiles.

Figure 23.1 - The General Mechanism of a Carbonyl Condensation Reaction



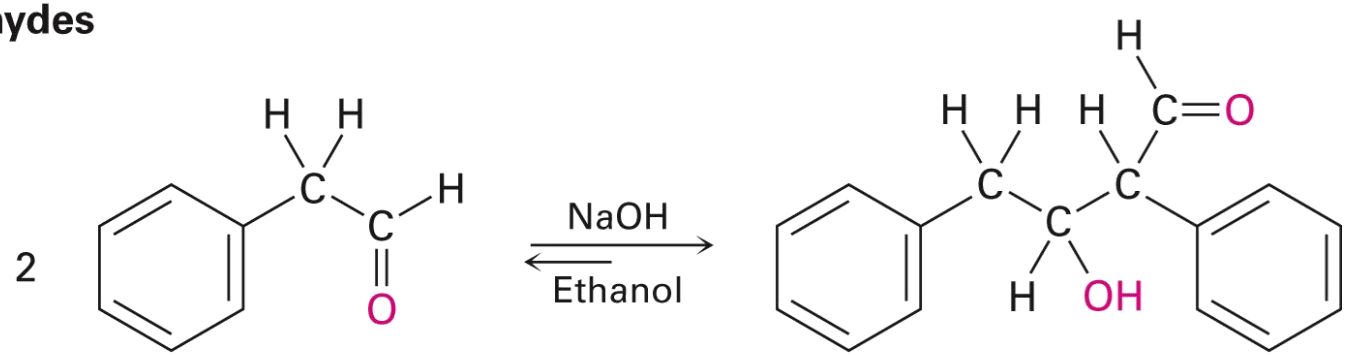
Carbonyl Condensations: The Aldol Reaction

- An aldol reaction is a base-catalysed carbonyl condensation reaction that occurs in aldehydes and ketones with an α hydrogen atom
  - The term aldol comes from the product formed, which contains an aldehyde and an alcohol



Aldehydes and Ketones and the Aldol Equilibrium

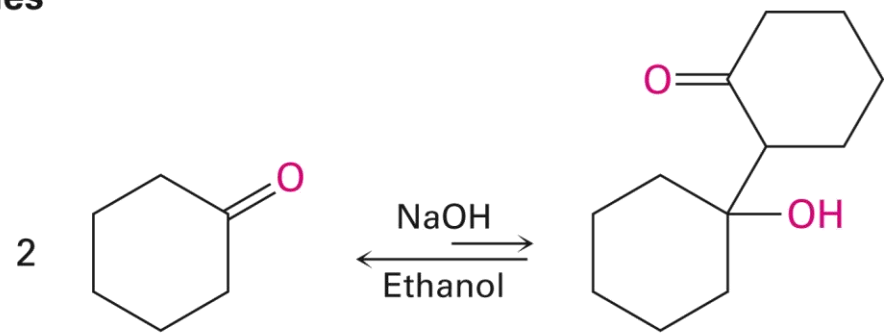
Aldehydes



Phenylacetaldehyde (10%)

(90%)

Ketones

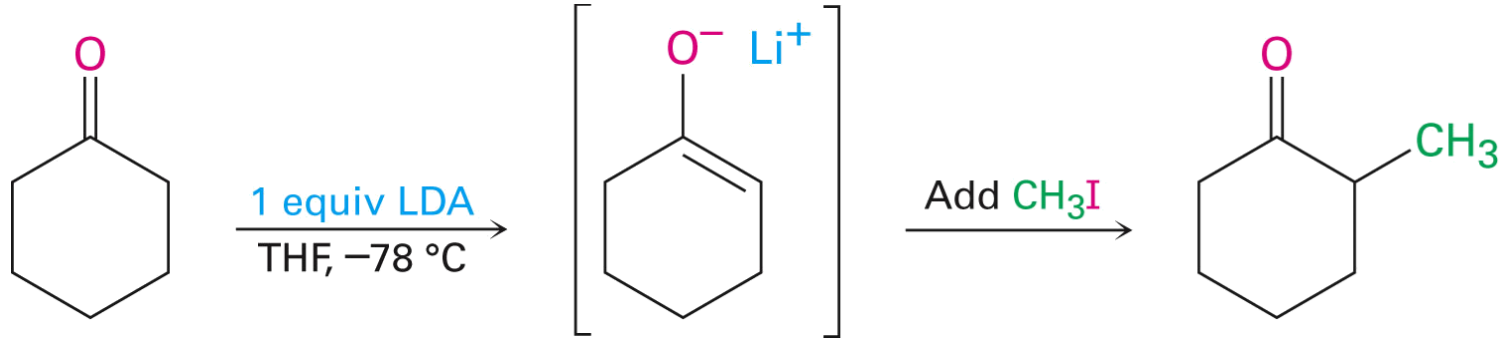


Cyclohexanone (78%)

(22%)

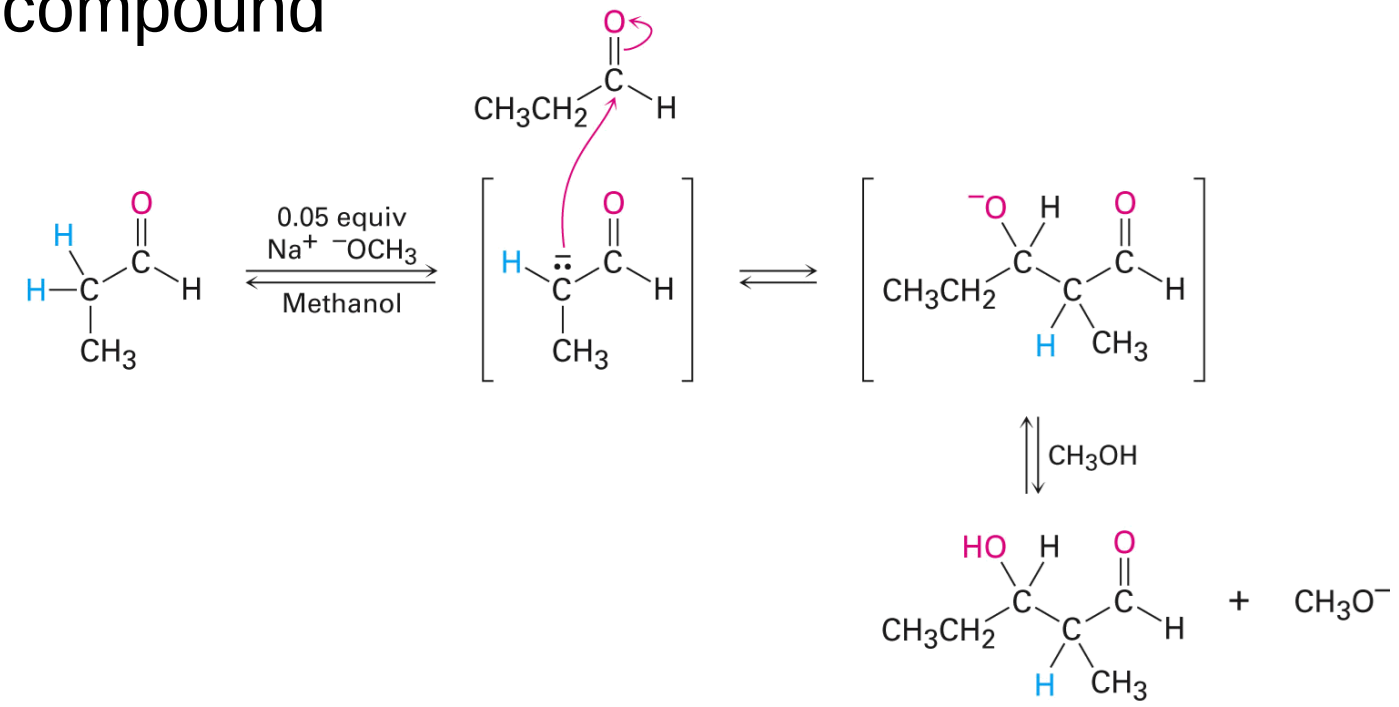
Carbonyl Condensations versus Alpha Substitutions

- Carbonyl condensations and α substitutions have similar experimental conditions
  - Control of experimental conditions is necessary in determining the outcome of a reaction
- In an α-substitution reaction, a strong base used and the conditions created ensure that the carbonyl compound quickly converts to an enolate ion at a low temperature



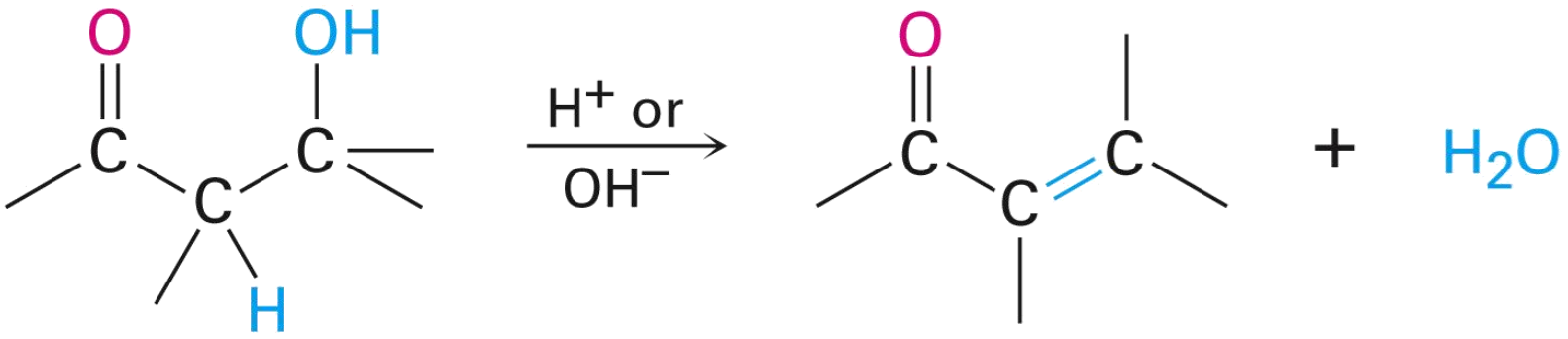
Carbonyl Condensations versus Alpha Substitutions

- In a carbonyl condensation, a catalytic amount of a relatively weak base is required
  - Result is the formation of a small amount of enolate ion in the presence of unreacted carbonyl compound



Dehydration of Aldol Products: Synthesis of Enones

- The β-hydroxy carbonyl products dehydrate to yield conjugated enones
- The term condensation is based on the fact that that water condenses during the formation of an enome product

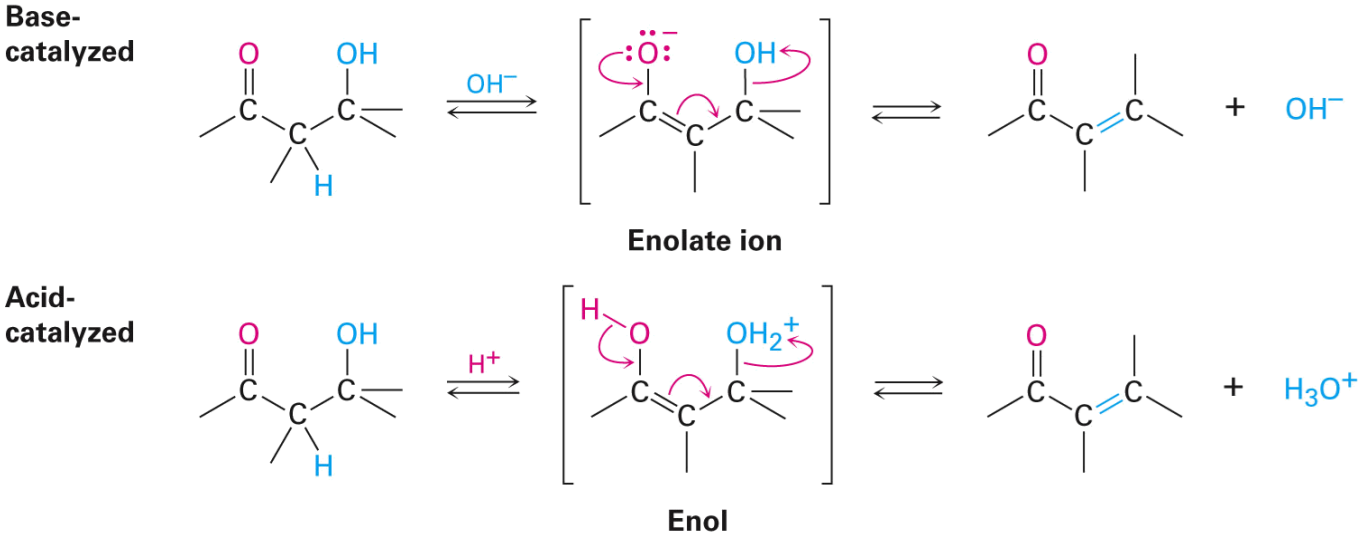


A β-hydroxy ketone or aldehyde

A conjugated enone

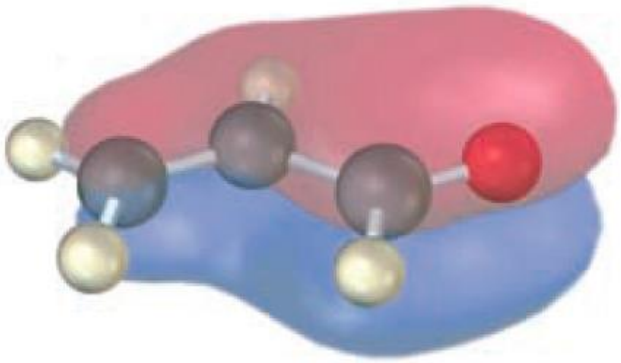
Dehydration of Aldol Products: Synthesis of Enones

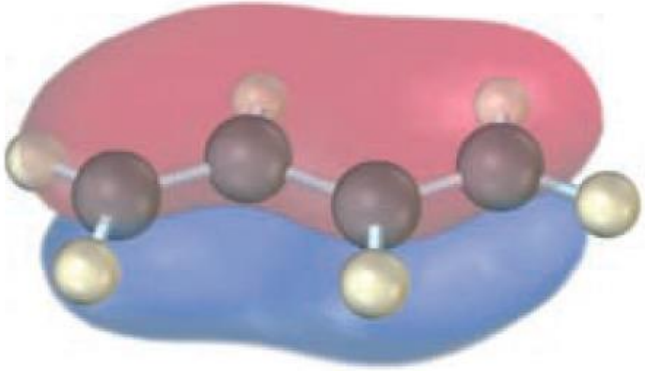
- Dehydration in aldol products is due to their carbonyl group
  - The α hydrogen is removed by a base yielding an enolate ion that expels the –OH leaving group
  - In acidic conditions, the –OH group is protonated and water is expelled

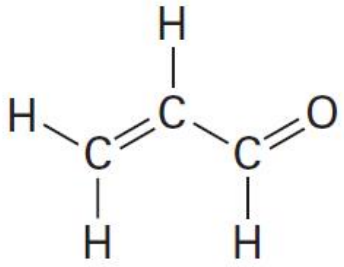


Dehydration of Aldol Products: Synthesis of Enones

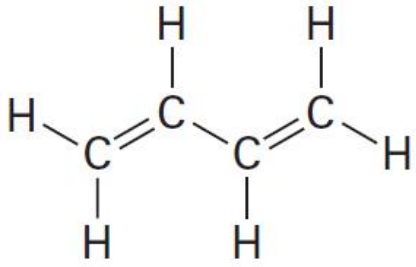
- Conjugated enones are more stable than nonconjugated enones







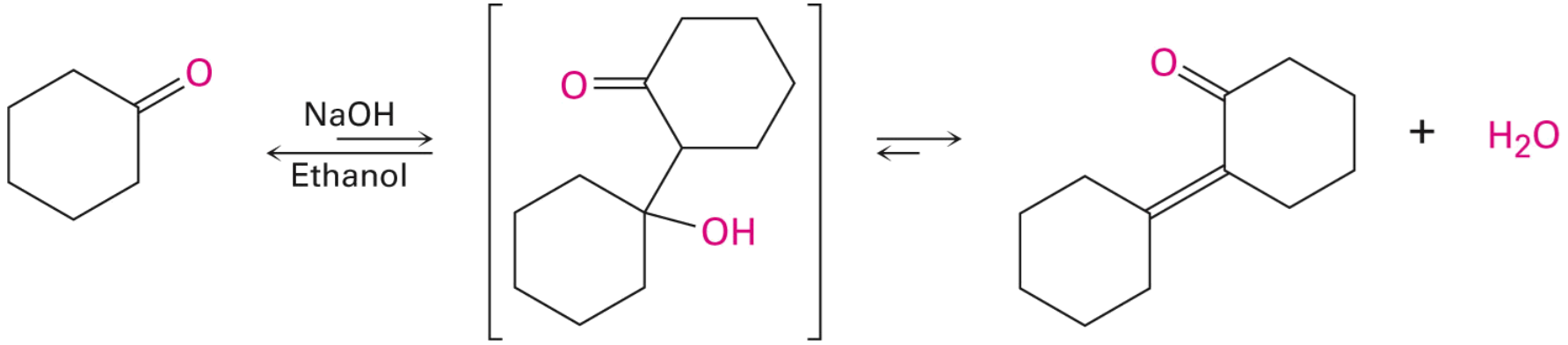
Propenal



1,3-Butadiene

Controlling Aldol Equilibrium

- Removal of water from the aldol reaction mixture can be used to drive the aldol equilibrium toward the product
- Despite the unfavourable initial aldol step, the following dehydration step results in the many aldol condensations

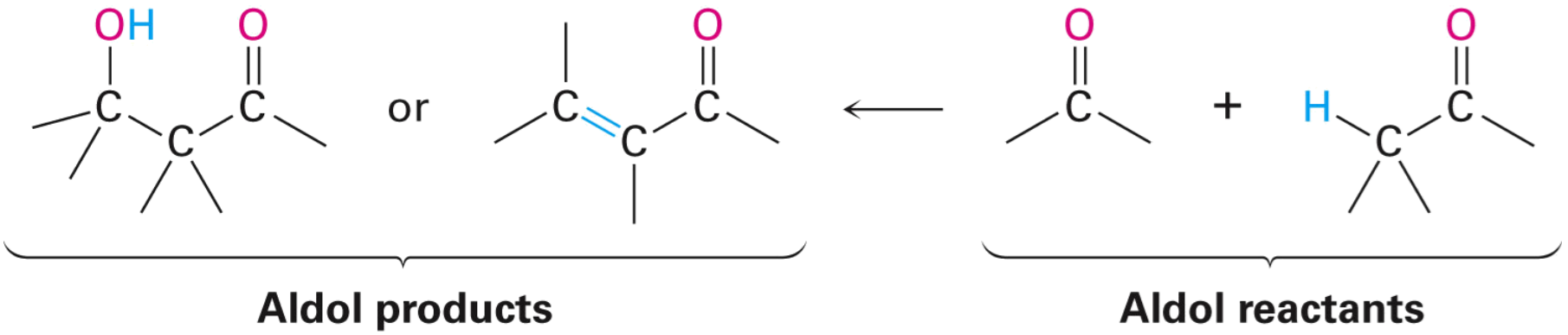


Cyclohexanone

Cyclohexylidenecyclohexanone (92%)

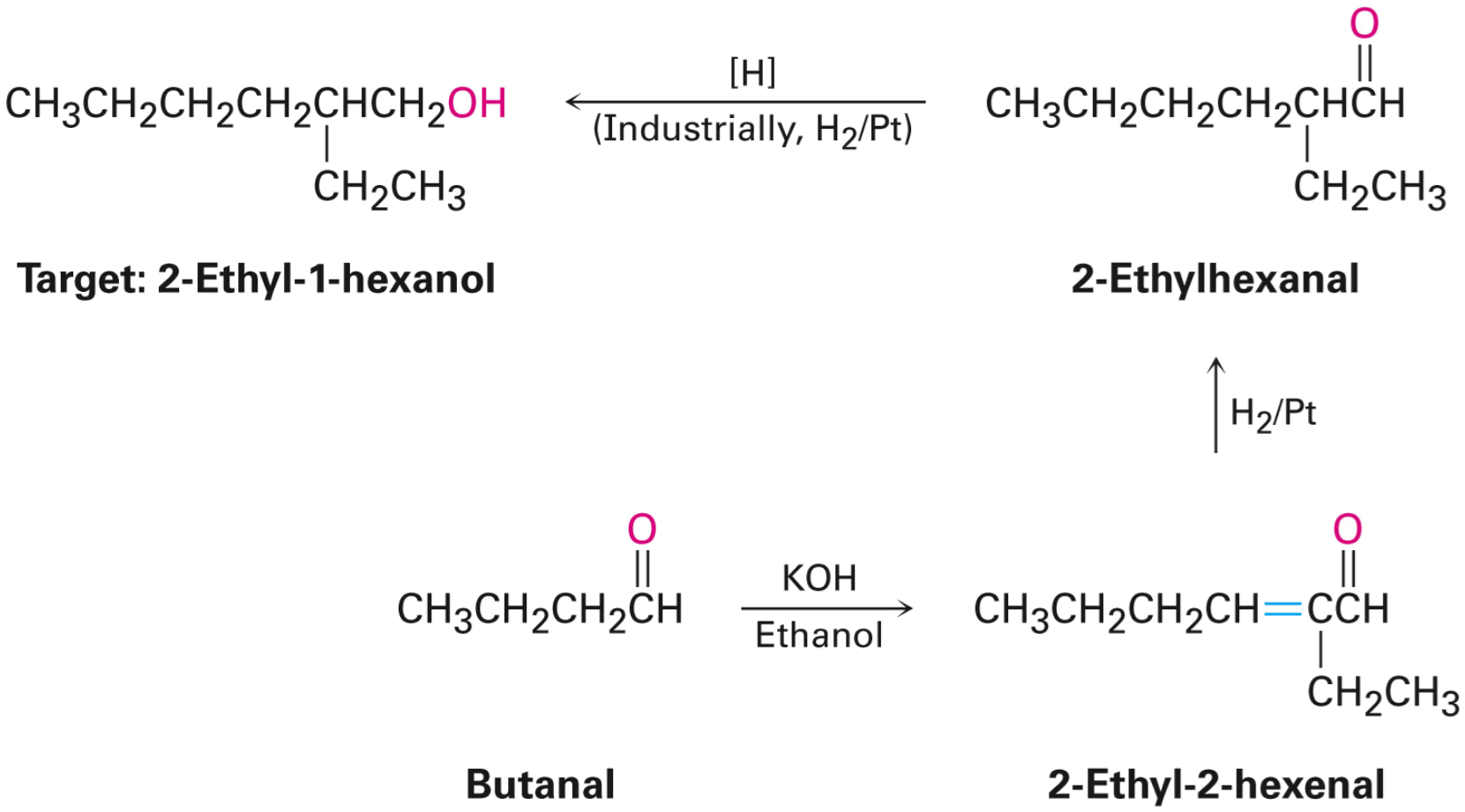
Using Aldol Reactions in Synthesis

- Experimental conditions determine the product of the aldol reaction
  - Working backwards may help to foresee when the aldol reaction may be useful in synthesis
- If a desired molecule contains either a β-hydroxy carbonyl or a conjugated enone, it might come from an aldol reaction



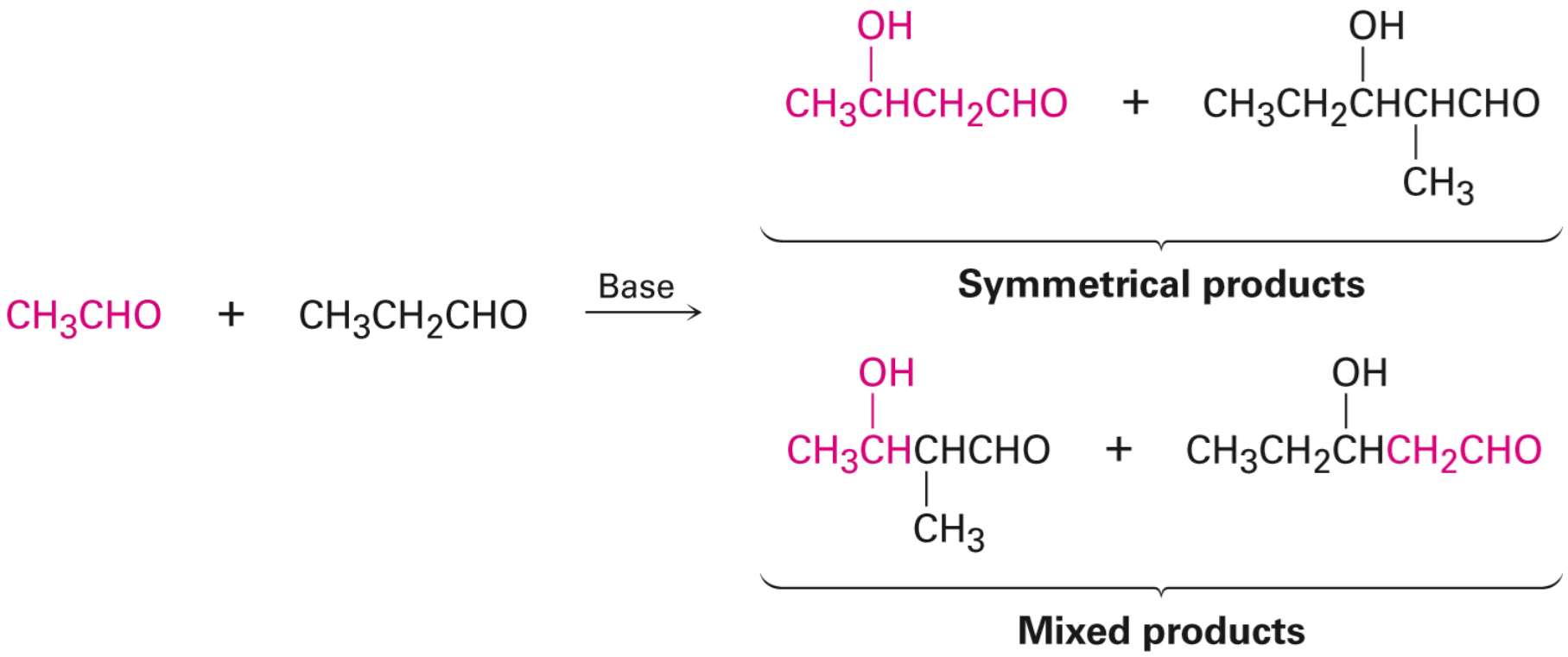
Extending the Synthesis

- Subsequent transformations can be carried out on the aldol products



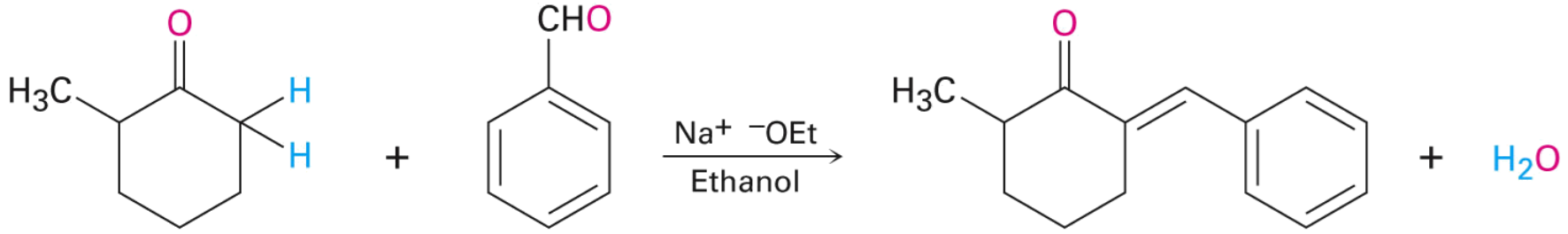
Mixed Aldol Reactions

- A mixed aldol reaction between two similar aldehyde or ketone partners leads to a mixture of four possible products



Practical Mixed Aldols

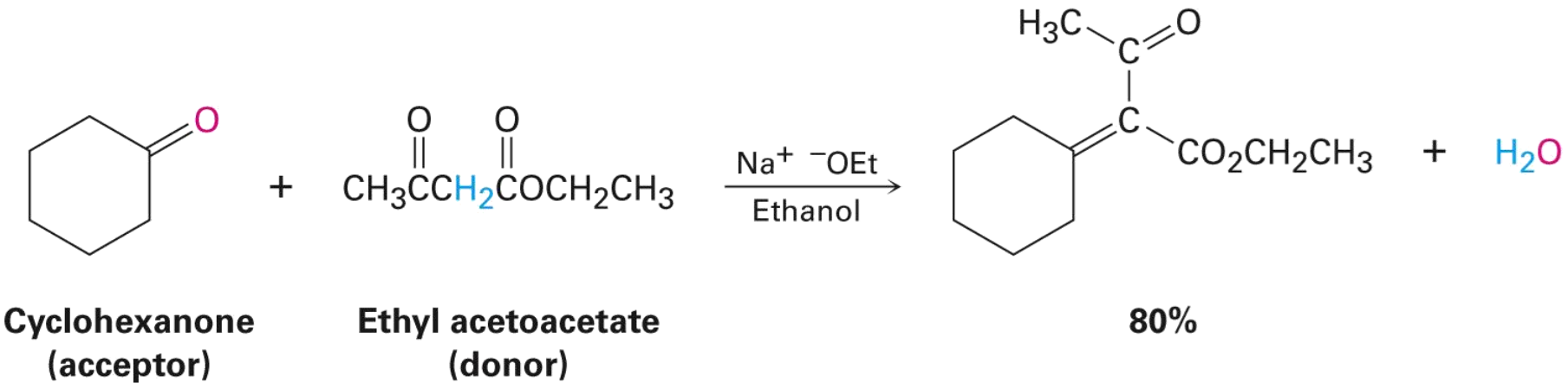
- A mixed aldol reaction is possible when:
  - One of the carbonyl partners have no α hydrogens
  - An unhindered carbonyl group is present



2-Methylcyclohexanone (donor) **Benzaldehyde (acceptor)** **78%**

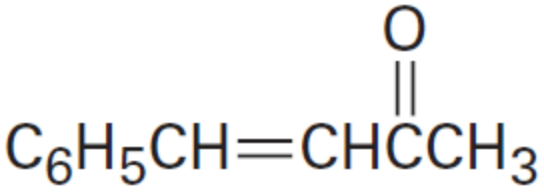
Practical Mixed Aldols

- A successful mixed aldol reaction is also possible when one of the carbonyl partners is transformed into its enolate ion on account of a higher acidic nature than the other carbonyl partner



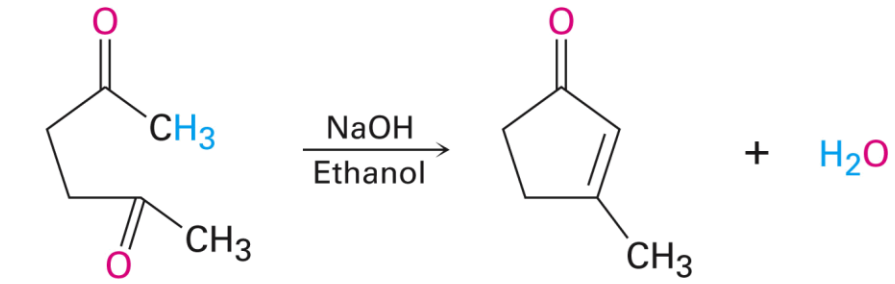
Practical Mixed Aldols

- Determine if the following compound can be prepared by a mixed aldol reaction
  - Show the reactant that can be used

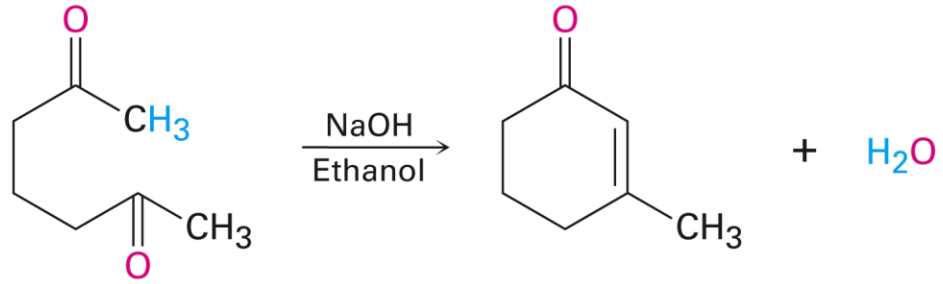


Intramolecular Aldol Reactions

- Treatment of certain dicarbonyl compounds with base produces cyclic products by intramolecular reaction

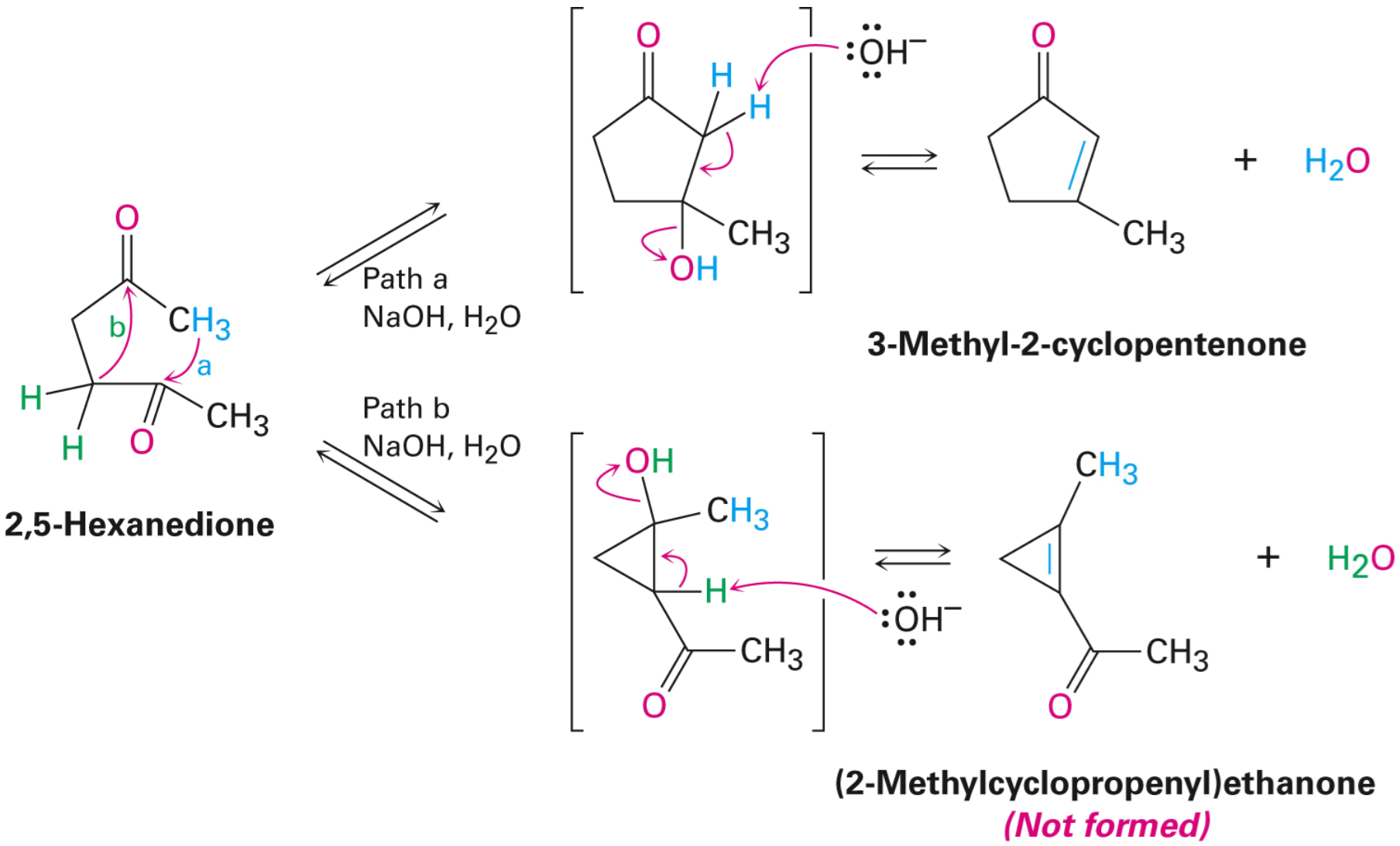


2,5-Hexanedione (a 1,4-diketone) **3-Methyl-2-cyclopentenone**



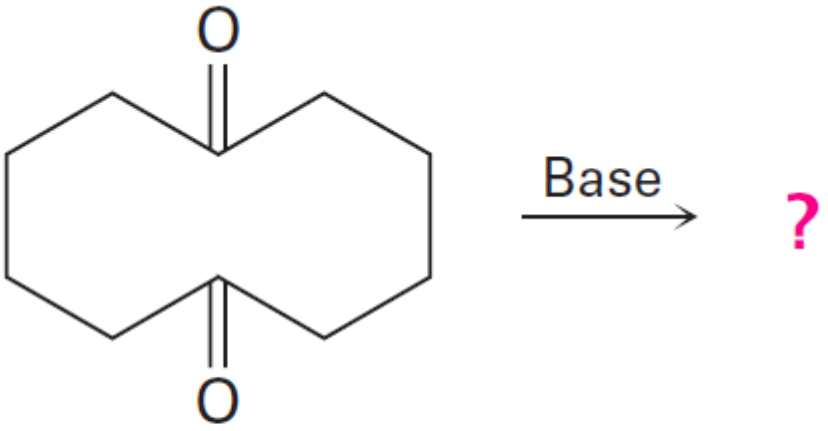
2,6-Heptanedione (a 1,5-diketone) **3-Methyl-2-cyclohexenone**

Figure 23.3 - Intramolecular Aldol Reaction of 2,5-hexanedione



Worked Example

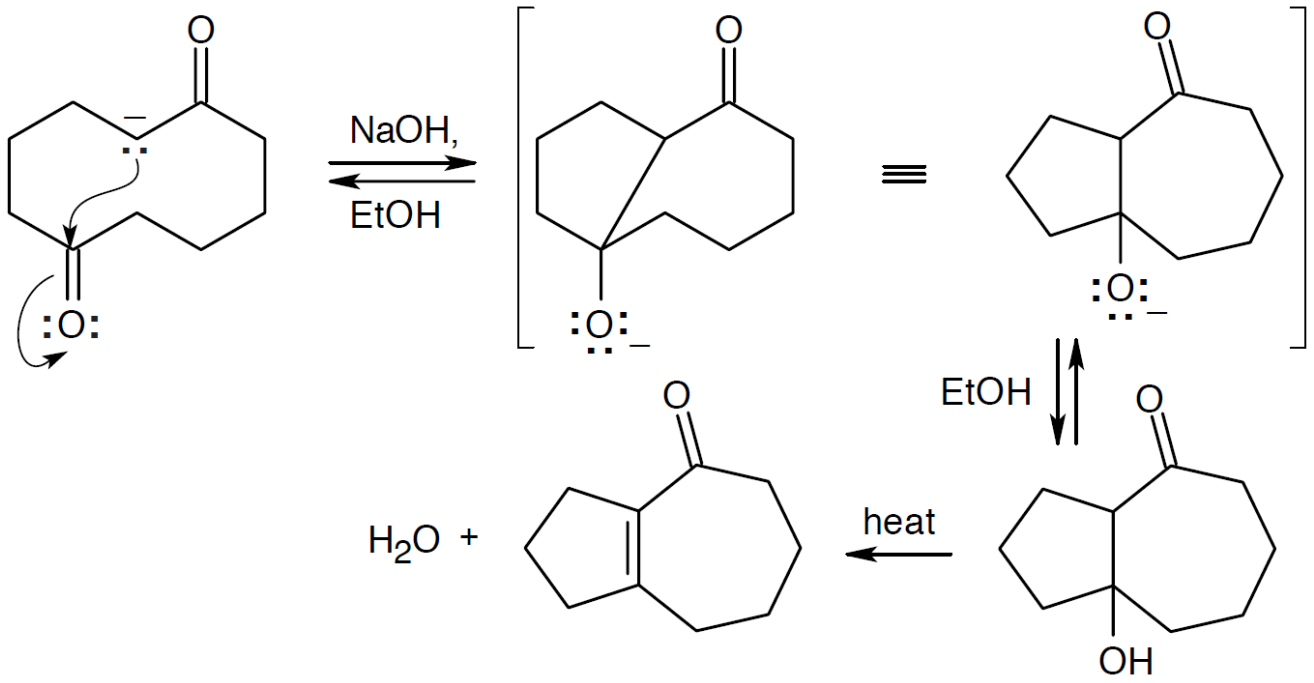
- Determine the product from base treatment of 1,6-cyclodecanedione



1,6-Cyclodecanedione

Worked Example

- Solution:
  - This intramolecular aldol condensation gives a product with a seven-membered ring fused to a five-membered ring



Claisen Condensation Reaction

- Occurs between two ester molecules
  - An ester with an α hydrogen undergoes reversible carbonyl condensation reaction when treated with 1 equivalent of a base resulting in a β-keto ester

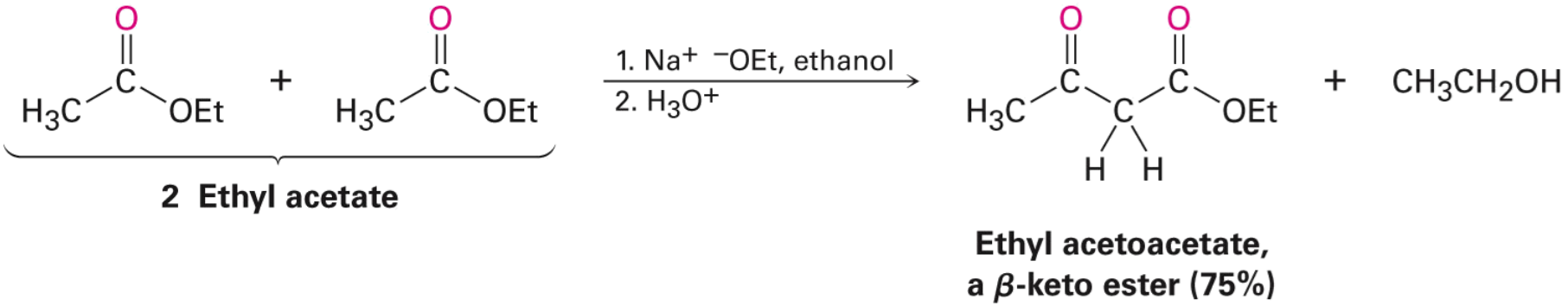
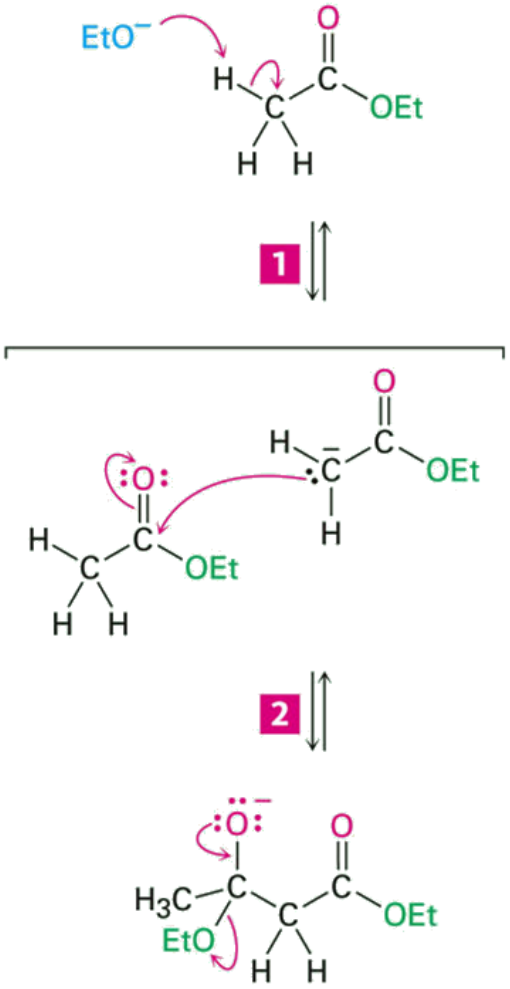


Figure 23.4 - Mechanism of the Claisen Condensation Reaction



1 Base abstracts an acidic alpha hydrogen atom from an ester molecule, yielding an ester enolate ion.

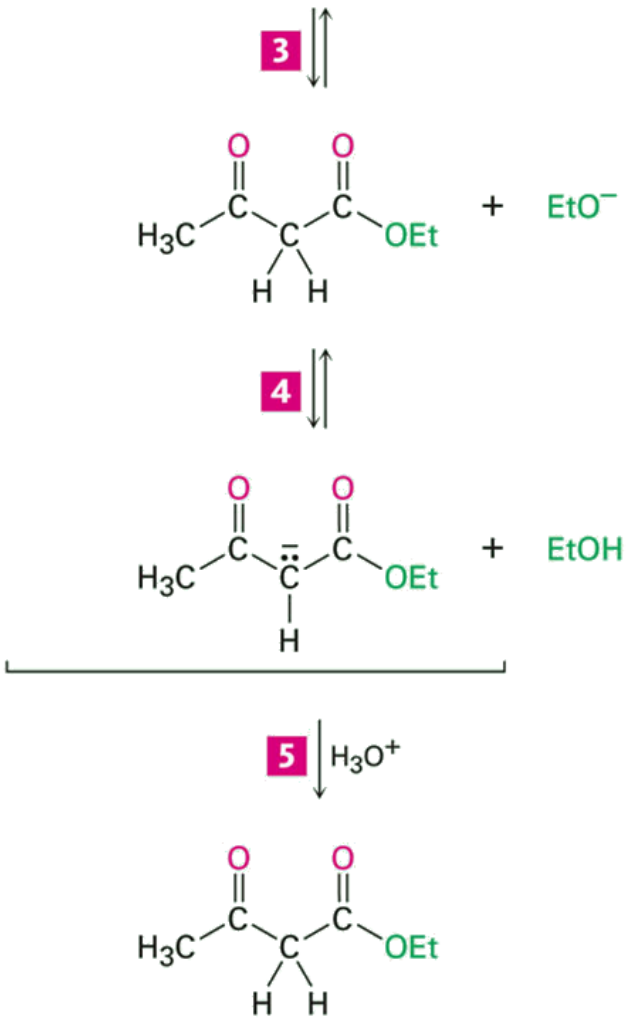
2 The enolate ion adds in a nucleophilic addition reaction to a second ester molecule, giving a tetrahedral alkoxide intermediate.

Figure 23.4 - Mechanism of the Claisen Condensation Reaction

3 The tetrahedral intermediate expels ethoxide ion to yield a new carbonyl compound, ethyl acetoacetate.

4 But ethoxide ion is a strong enough base to deprotonate ethyl acetoacetate, shifting the equilibrium and driving the overall reaction to completion.

5 Protonation of the enolate ion by addition of aqueous acid in a separate step yields the final β-keto ester product.

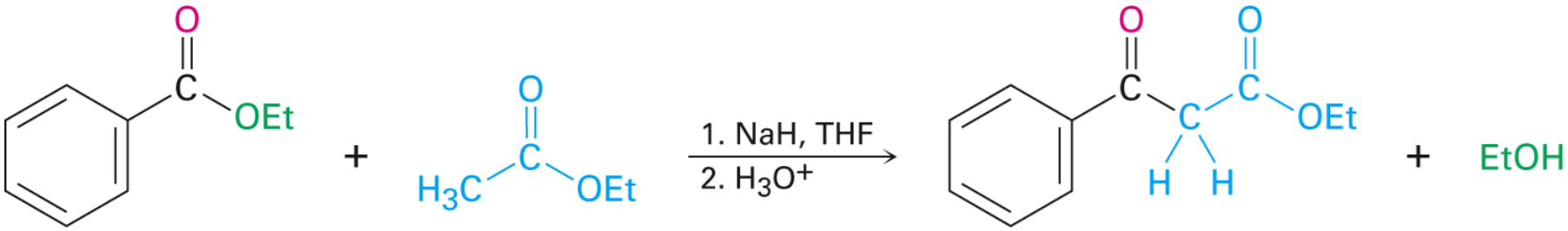


Features of the Claisen Condensation

- If the starting ester has more than one acidic α hydrogen, the product β-keto ester has a doubly activated hydrogen atom that can be abstracted by base
  - Requires a full equivalent of base rather than a catalytic amount
- Deprotonation of the product causes a shift in equilibrium over to the side of the product, resulting in high yields

Mixed Claisen Condensations

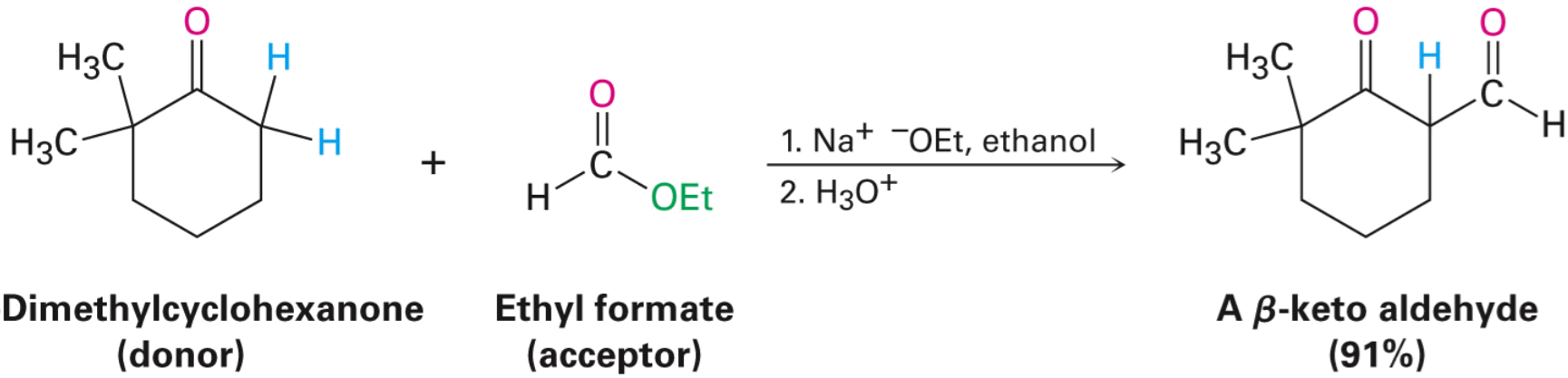
- Successful when one of the two esters cannot form an enolate ion as it does not possess α hydrogens



Ethyl benzoate (acceptor) + **Ethyl acetate (donor)** → **Ethyl benzoylacetate** + EtOH

Causing Claisen-like Reactions in Esters and Ketones

- Reactions between esters and ketones result in β-diketones
- An ester component with no α hydrogens is favorable



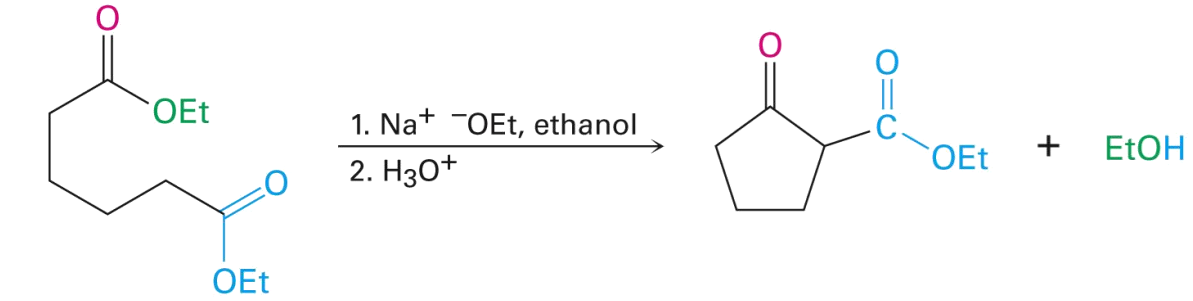
2,2-Dimethylcyclohexanone (donor)

Ethyl formate (acceptor)

A β-keto aldehyde (91%)

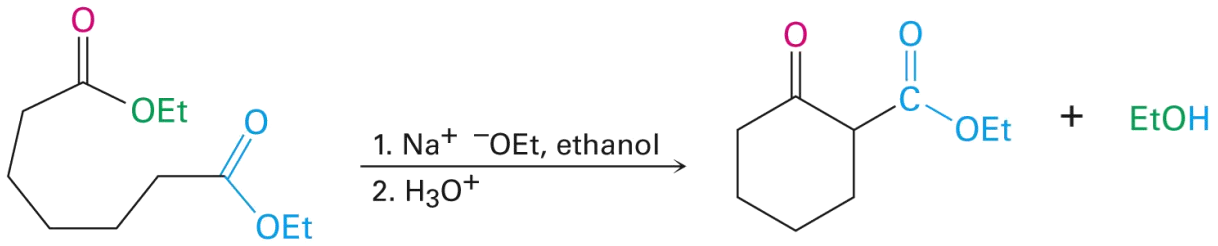
Intramolecular Claisen Condensations: The Dieckmann Cyclization

- It can occur in diketones
- Effective **Dieckmann cyclization** reactions occur in 1,6-diesters and 1,7-diesters



Diethyl hexanedioate (a 1,6-diester)

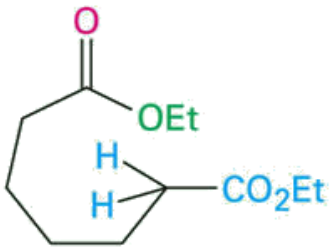
Ethyl 2-oxocyclopentanecarboxylate (82%)



Diethyl heptanedioate (a 1,7-diester)

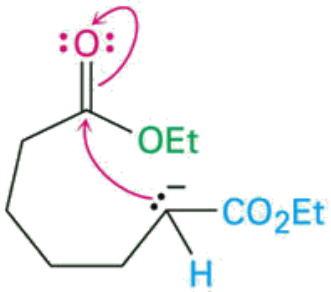
Ethyl 2-oxocyclohexanecarboxylate

Figure 13.5 - Mechanism of the Dieckmann Cyclization



1 Base abstracts an acidic α proton from the carbon atom next to one of the ester groups, yielding an enolate ion.

1 $\rightleftharpoons$ $Na^+ {}^-OEt$



+ EtOH

2 Intramolecular nucleophilic addition of the ester enolate ion to the carbonyl group of the second ester at the other end of the chain then gives a cyclic tetrahedral intermediate.

2 $\rightleftharpoons$

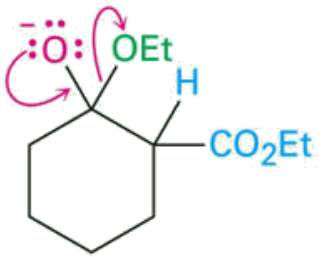
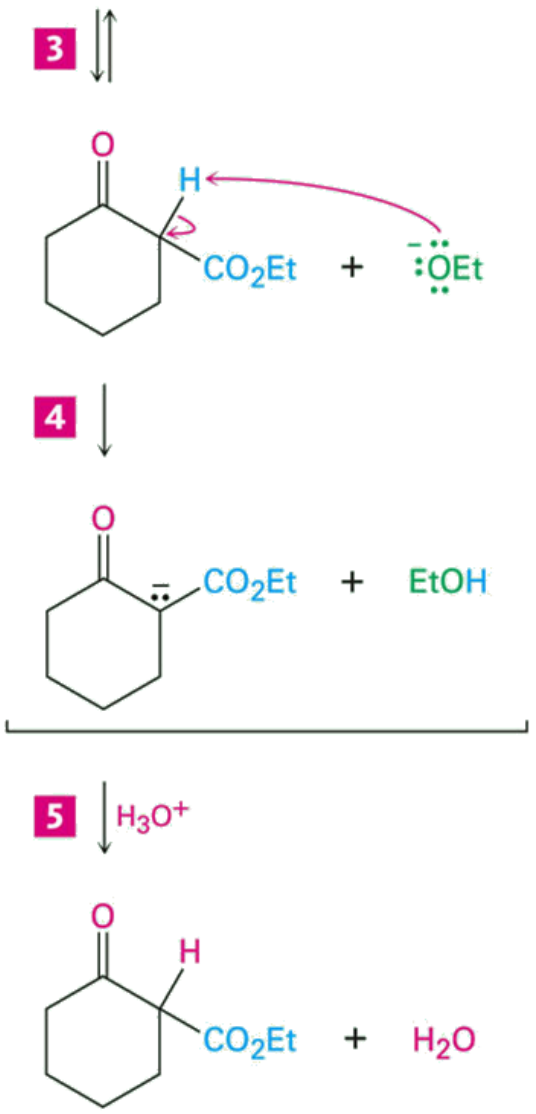


Figure 13.5 - Mechanism of the Dieckmann Cyclization

3 Loss of alkoxide ion from the tetrahedral intermediate forms a cyclic β-keto ester.

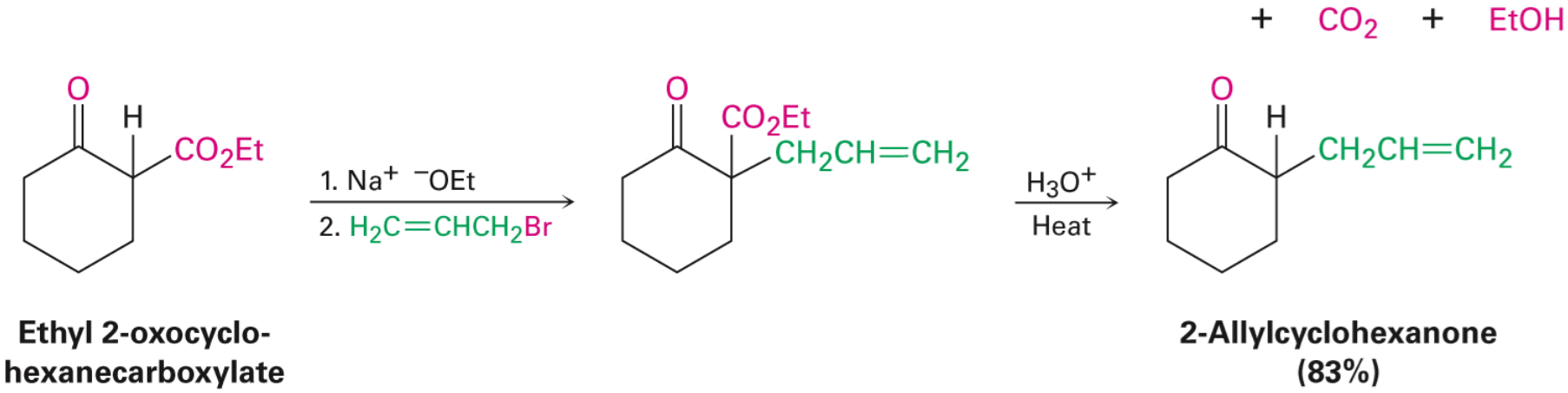
4 Deprotonation of the acidic β-keto ester gives an enolate ion . . .

5 . . . which is protonated by addition of aqueous acid at the end of the reaction to generate the neutral β-keto ester product.



Alkylating the Product of the Dieckmann Reaction

- The cyclic β-keto ester can be further alkylated and decarboxylated as in the acetoacetic ester synthesis

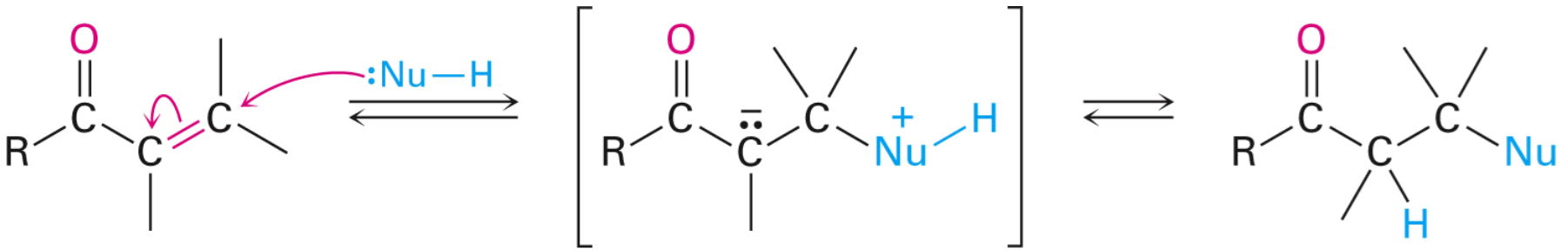


Ethyl 2-oxocyclo-hexanecarboxylate

2-Allylcyclohexanone (83%)

Conjugate Carbonyl Additions: The Michael Reaction

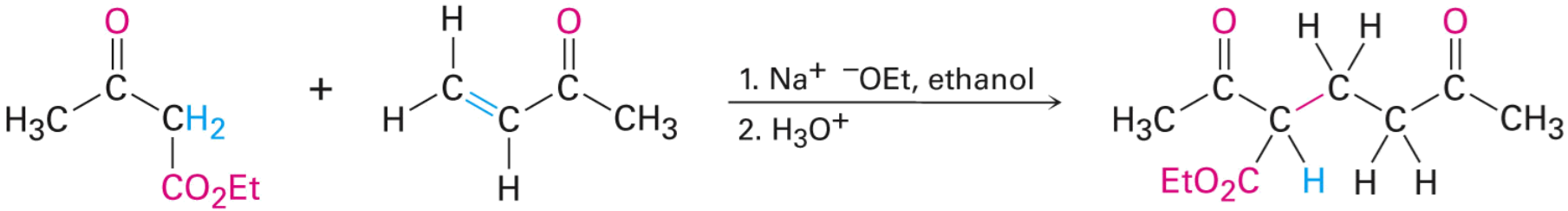
- Process in which a nucleophilic enolate ion reacts with an α,β-unsaturated compound, producing a conjugate addition product



Conjugate addition product

Optimal Conditions for the Michael Reaction

- Involve a particularly stable ion and an unhindered α,β-unsaturated ketone



Ethyl acetoacetate

3-Buten-2-one

Figure 23.6 - Mechanism of the Michael Reaction

1 The base catalyst removes an acidic alpha proton from the starting β-keto ester to generate a stabilized enolate ion nucleophile.

2 The nucleophile adds to the α,β-unsaturated ketone electrophile in a Michael reaction to generate a new enolate as product.

3 The enolate product abstracts an acidic proton, either from solvent or from starting keto ester, to yield the final addition product.

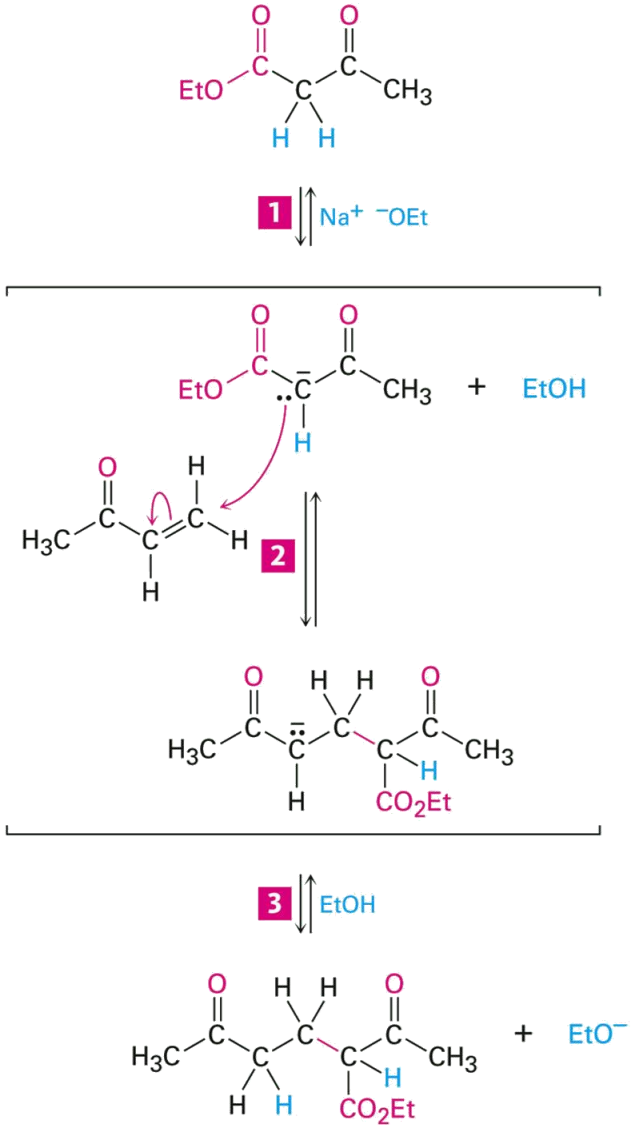
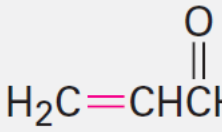
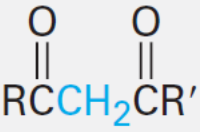
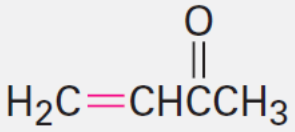
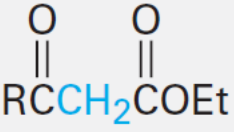
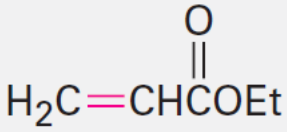
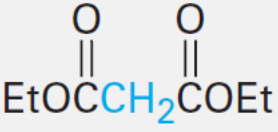
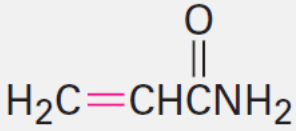
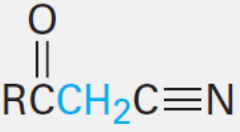
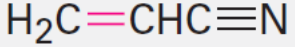
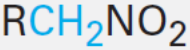
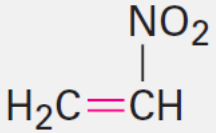
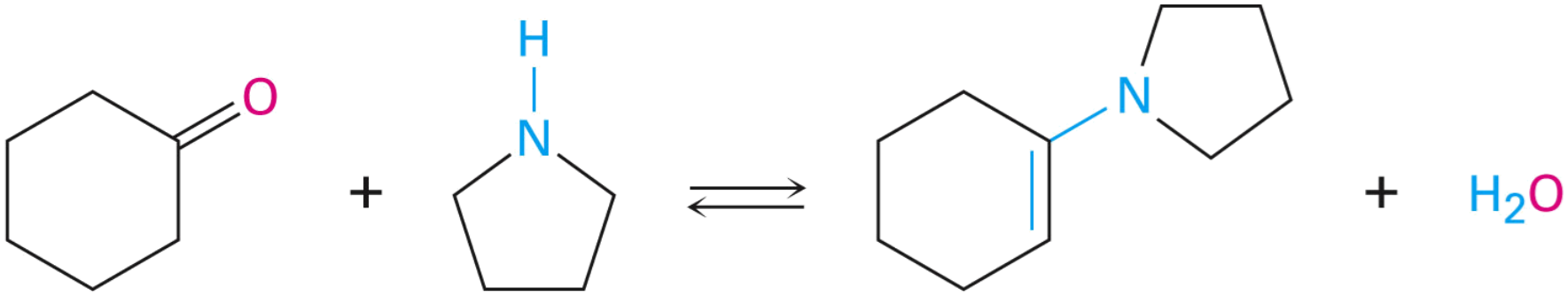


Table 23.1 - Some Michael Acceptors and Some Michael Donors

Michael acceptors		Michael donors	
	Propenal		β-Diketone
	3-Buten-2-one		β-Keto ester
	Ethyl propenoate		Diethyl malonate
	Propenamide		β-Keto nitrile
	Propenenitrile		Nitro compound
	Nitroethylene		

Carbonyl Condensations with Enamines: The Stork Reaction

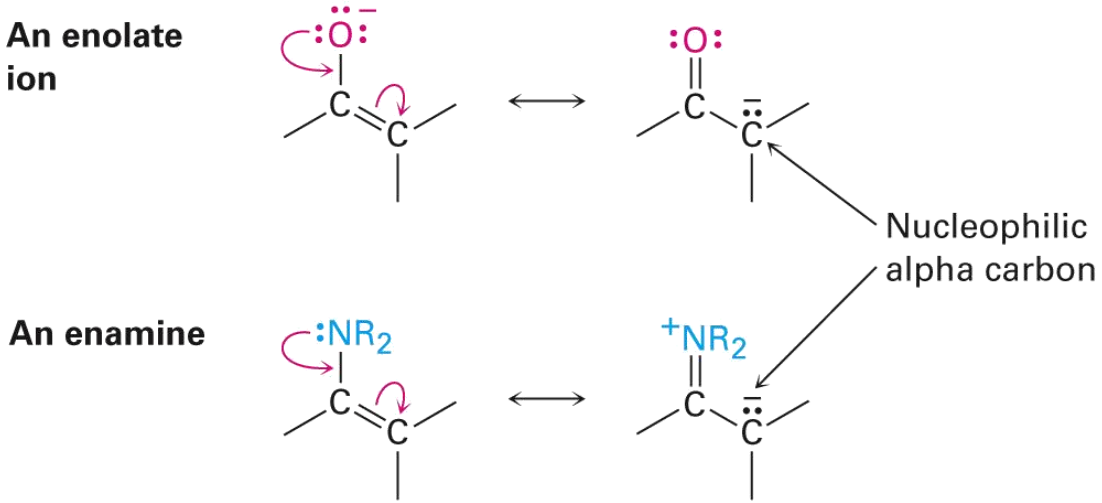
- Enamines are prepared from a ketone and a secondary amine

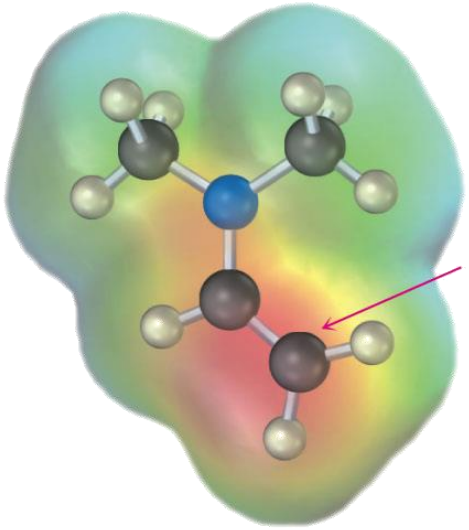


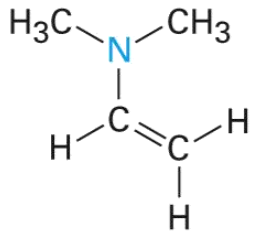
Cyclohexanone **Pyrrolidine** **1-Pyrrolidino-cyclohexene (87%)**

Enamines Are Nucleophilic

- Overlap of the nitrogen lone-pair orbital with the double-bond π orbitals increases electron density on the α carbon atom

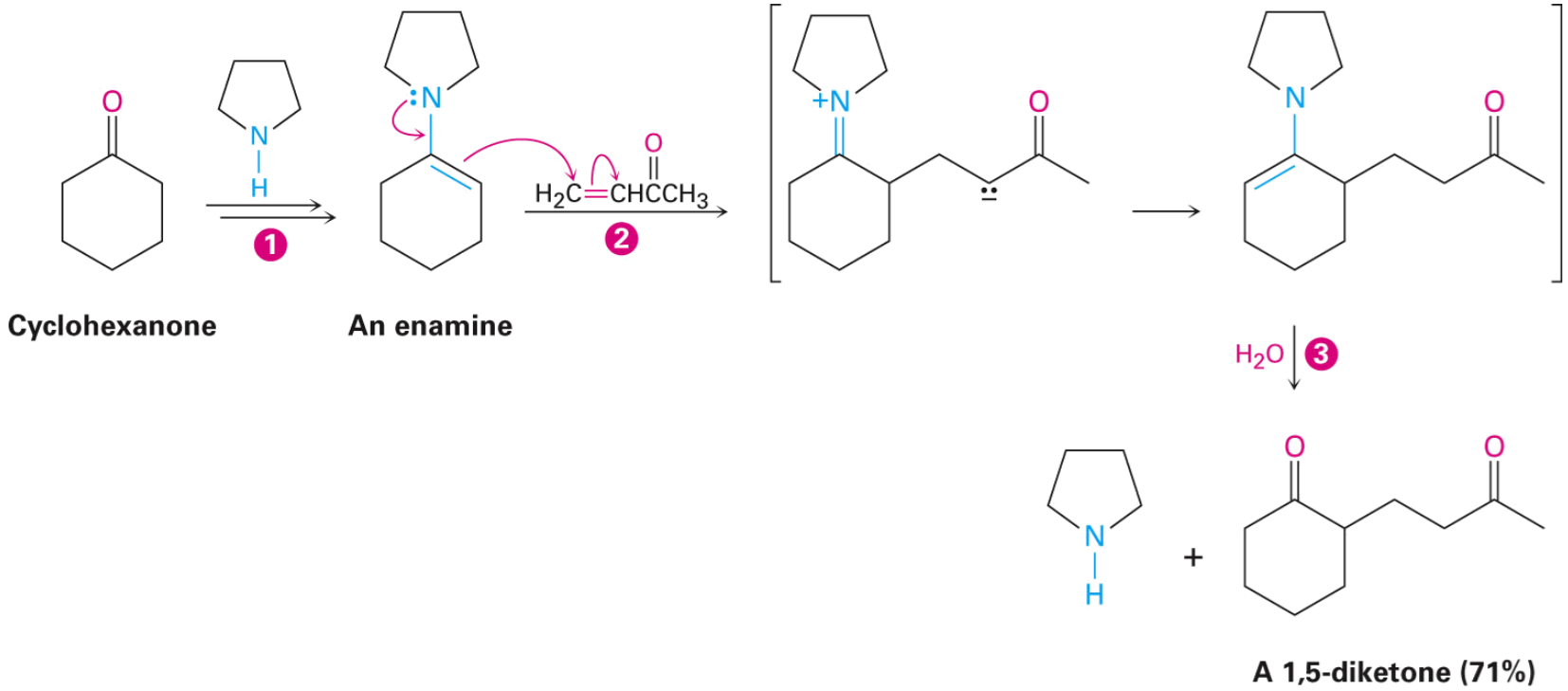






Enamine Addition and Hydrolysis

- Enamine adds to an α,β-unsaturated carbonyl acceptor
- Product is hydrolyzed to a 1,5-dicarbonyl compound



The Robinson Annulation Reaction

- A two-step process
  - It combines a Michael reaction with an intramolecular aldol reaction
- The reactants required are a nucleophilic donor, an enamine, and an α,β-unsaturated ketone acceptor

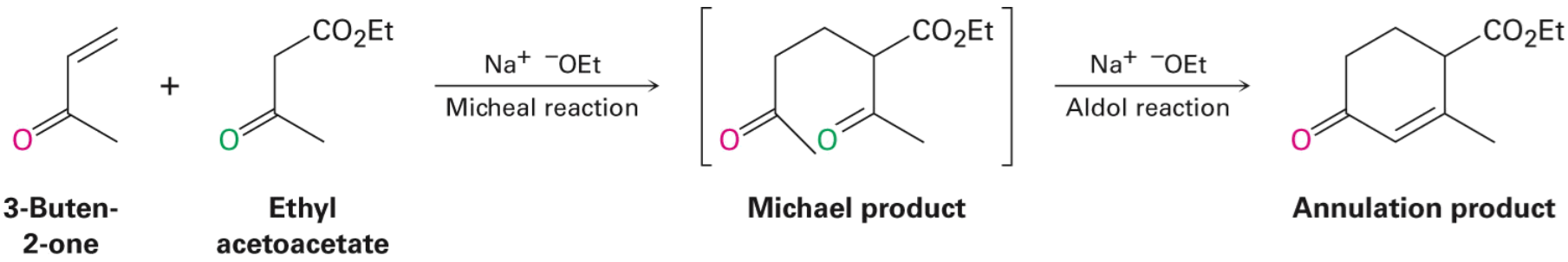
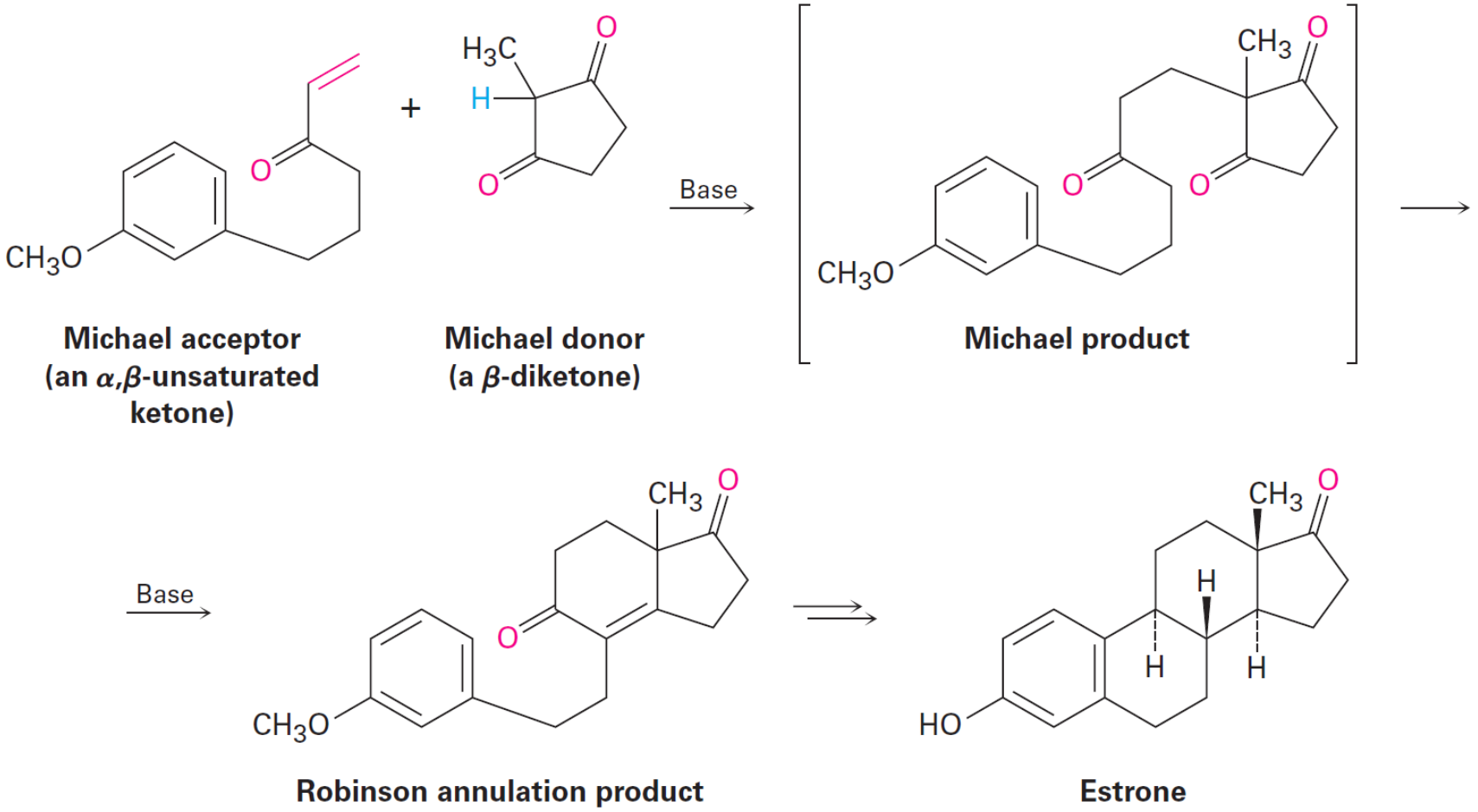
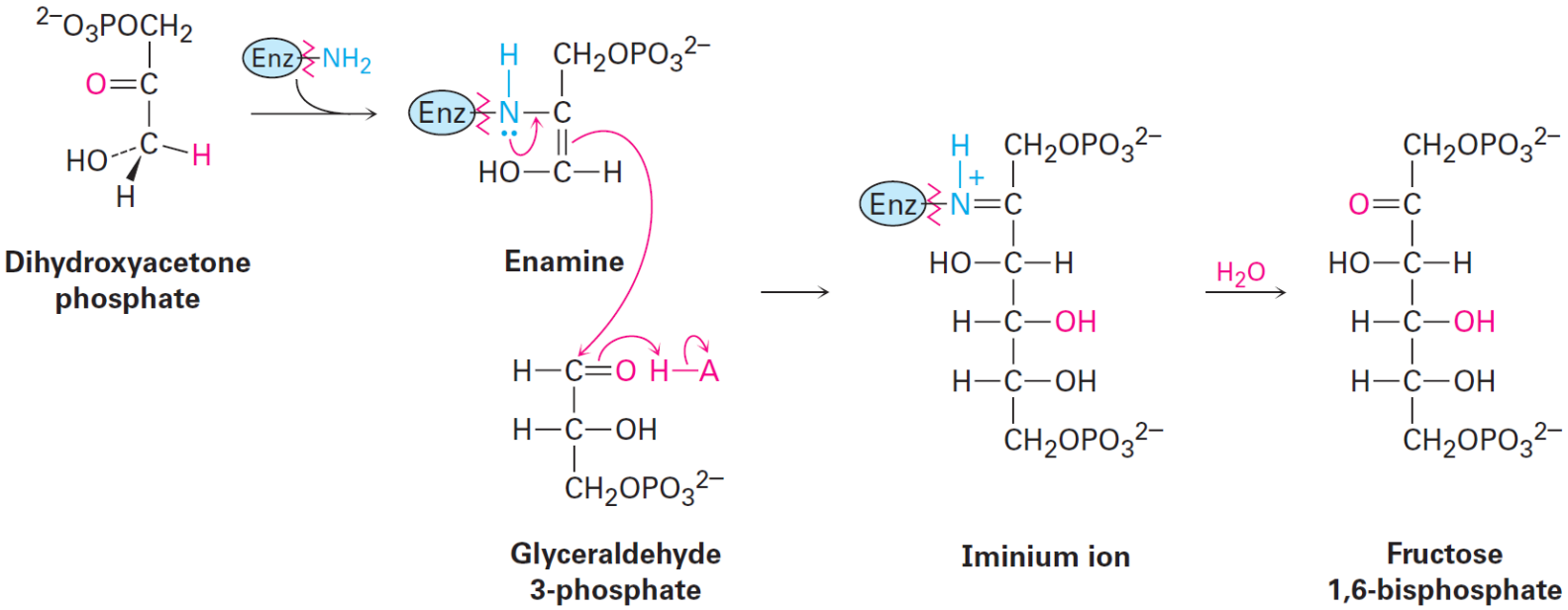


Figure 23.8 - Synthesis of the Steroid Hormone Estrone



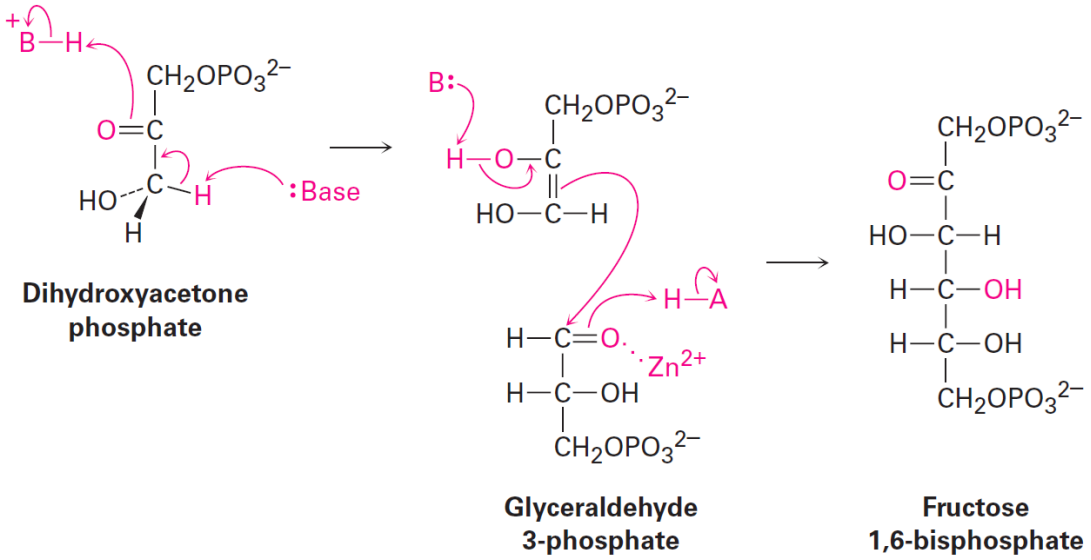
Some Biological Carbonyl Condensation Reactions

- Biological aldol reactions are widely found in carbohydrate metabolism
  - Type I aldolases are present mostly in animals and higher plants



Some Biological Carbonyl Condensation Reactions

- Type II aldolases are present mostly in fungi and bacteria



- Aldolase-catalysed reactions are mixed aldol reactions that occur between two different partners

Biological Claisen Condensations

- They are prevalent in many biological pathways
  - Fatty acid biosynthesis

